

⁵³Cr (Cr I), nuclear spin $I = 3/2$. Strongest classified transitions below 2 μm (NIST): 56 levels, 87 lines.

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J J\rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
a ¹ S ₀ - y ¹ P ₀ ^o	357.9708	357.8666	837.4775	5.565	1.520e+08	24.11	100	strong UV resonance line a753 - y7P4
a ¹ S ₀ - y ¹ P ₂ ^o	359.4511	359.3485	834.0285	4.922	1.510e+08	24.11	99.7	strong UV resonance line a753 - y7P3
a ¹ S ₀ - y ¹ P ₂ ^o	360.6357	360.5329	831.2888	4.181	1.510e+08	24.11	99.7	strong UV resonance line a753 - y7P2
a ¹ S ₀ - z ¹ P ₀ ^o	425.5534	425.4336	704.4768	3.361	3.300e+07	5.253	100	main cooling line: Zeeman slower, MOT, imaging
a ¹ S ₀ - z ¹ P ₃ ^o	427.6000	427.4797	701.1049	2.926	3.170e+07	5.053	99.9	optical pumping to m _J = -3 and "depumper" towards 5S2 /...
a ¹ S ₀ - z ¹ P ₂ ^o	429.0924	428.9717	698.6664	2.458	3.100e+07	4.943	99.8	429.1 nm line
a ¹ S ₀ - z ¹ P ₀ ^o	633.1842	633.0091	473.4680	0.160	2.900e+04	5.053	0.0914	633.2 nm repumper 5S2 -> 7P3
a ³ D ₂ - z ¹ P ₀ ^o	646.8765	646.6978	463.4462	-	-	5.053	-	646.9 nm repumper 5D2 -> 7P3
a ³ D ₃ - z ¹ P ₀ ^o	649.1976	649.0182	461.7892	0.007	4.200e+01	5.253	0.000127	intercombination leak of the cooling line into metastable...
a ³ D ₃ - z ¹ P ₃ ^o	653.9727	653.7921	458.4174	-	-	5.053	-	654.0 nm repumper 5D3 -> 7P3
a ³ D ₄ - z ¹ P ₀ ^o	658.2741	658.0923	455.4219	0.013	1.270e+02	5.253	0.000385	intercombination leak of the cooling line into metastable...
a ³ D ₄ - z ¹ P ₃ ^o	663.1842	663.0011	452.0500	0.078	6.000e+03	5.053	0.0189	663.2 nm repumper 5D4 -> 7P3

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)					
a ¹ S _{3/2}	-83.5990(20)	-	F=3/2: +701.59	F=5/2: +292.60	F=7/2: -0.00	F=9/2: -376.20		
a ¹ S _{5/2}	-173.820	-	F=1/2: -782.18	F=3/2: +521.46	F=5/2: +86.91	F=7/2: -521.46		
a ¹ D ₂	-35.683(5)	15.565(5)	F=3/2: +223.44	F=5/2: +122.56	F=7/2: -7.78	F=9/2: -156.68		
a ¹ D ₄	-48.755(5)	63.021(10)	F=5/2: +396.61	F=7/2: +182.64	F=9/2: -51.95	F=11/2: -276.77		
z ¹ P _{3/2}	-1.5(20)	-2.000	F=3/2: +7.80	F=5/2: +5.55	F=7/2: +1.00	F=9/2: -7.25		
z ¹ P _{1/2}	-11.8(4)	-0.6(6)	F=5/2: +88.20	F=7/2: +47.32	F=9/2: -5.64	F=11/2: -70.95		

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^7S_3 - z^7P_4^o$	425.553	53-52	10.7(17)
$a^7S_3 - z^7P_3^o$	427.600	53-52	18(23)
$a^5S_2 - z^7P_3^o$	633.184	53-52	-155(53)
$a^5D_3 - z^7P_3^o$	653.973	53-52	670(28)
$a^3D_4 - z^7P_3^o$	663.184	53-52	819(28)

Shift = ν (first isotope) - ν (second isotope).